An ATM network futures test bed

J W R Barnes, J Chalmers, P Cochrane, D Ginsburg, I D Henning, D J Newson and D J Pratt

The merging of communications and computing has spurred the development of broadband networking technologies. ATM (asynchronous transfer mode) has been proposed as a primary standard for combining different traffic types over a single bearer network to achieve a truly 'multiservice' capability. However, this technology is notoriously difficult to model and understand as it is not hierarchical nor well behaved, but more chaotic and self-organising. So a representative ATM-backboned network has been constructed — the 'futures test bed' — giving 700 people their own broadband connection, capable of providing a range of future services. The purpose of this test bed is to provide both a platform for the development of broadband applications, and a means for assessing the capabilities of ATM. This paper discusses the design and implementation of the network and considers some of the potential applications.

1. Introduction

Although the telecommunications industry is over 100 years old, most of the advance s in its technology have occurred over just the last fraction of its lifetime. The majority of today's networks are built with mature technologies developed within the last 20 years, and are well tested, characterised, and understood. However, exponentially increasing computing power, organisational change and fiercely competitive markets are accelerating the rate of change of network technology, services and operations. These factors are, in turn, increasing the uncertainty and risks associated with the deployment of new network technologies.

Asynchronous transfer mode (ATM) is a case in point. Traditional network technologies were designed to carry constant bit rate narrowband circuits over well-understood hierarchical topologies. But today's traffic is becoming increasingly unpredictable, bursty and broadband, so ATM has been developed to efficiently serve the requirements of both computers (bursty) and humans (real time). But at the dawn of the information age, people are more dependent upon their networks than ever before. In the information economy, reduced latency and increased bandwidth are becoming key sources of competitive advantage. In this context packet networks and data are very immature compared with circuit-switched networks and voice. With networks becoming flatter and their control more distributed, understanding of the technologies being deployed over the next decade is arguably more limited than ever before.

ATM is unusual in that its development has been rapid, and has involved not only telcos and their traditional equipment suppliers, but also computer and data networking manufacturers. Yet the requirements of these two communities are very different, and although such short development life cycles are becoming increasingly prevalent, it is currently believed that ATM is neither sufficiently mature nor well understood to offer the high reliability and high utilisation which characterise telco networks.

How then can this understanding be built-up? Valuable contributions are being made by computer modelling of ATM networks, but the real test comes in building them. A number of test bed networks have been established with multiple switches over large geographical areas, but here the alternative approach has been taken of building such a network for 700 users across a campus.

The 'futures test bed' is a live network which currently serves users at BT Laboratories with a combination of switched ethernet and direct ATM connections using equipment from Cisco Systems. These people are working at the frontiers of technology, from semiconductor device development and human factors to networked virtual reality and data visualisation. They are highly mobile, very interactive and are now totally dependent upon ATM in their daily work. They are also among the first to discover the daily benefits of a fast, highly interconnected network, and the limitations of an early operational ATM network.

This paper discusses the design and implementation of the futures test bed, the technical aspects of carrying TCP/IP over ATM using both the classical IP and LAN emulation standards, and concludes with a view of some of the applications and concerns which the test bed aims to address.

2. Test bed design and implementation

The spectrum of network requirements for the futures test bed matches most demandin g corporate environments. The users are scattered over 8 buildings separated by up to 1 km, and our design brief was to provide sufficient bandwidth to each user's desk for good quality video and interactive working applications.

Given that most of the users were already connected to ethernet networks, it was decided to continue to use ethernet at the edge, but to progressively upgrade by installing switching hubs. Fewer than 3-4 users per ethernet segment enables good quality video to be displayed on a PC equipped with a suitable MPEG decoder, but switched ethernet minimises the number of collisions to further improve quality at the price of slightly increased latency.

The technology choice for the network backbone was largely between FDDI (fibre distributed data interface) and ATM. Today, the performance of the network using either of these two technologies would be comparable. However, while FDDI's maturity is attractive, it does not offer clearly defined upgrade paths. In contrast ATM allows a core switch to be moved into the work-group as the core network speed is upgraded, thus reusing rather than discarding old switch equipment. This is quite separate from the reduced complexity of ATM compared with FDDI, which promises significant cost advantages for ATM in the future. ATM also offers shared paths for real time and data traffic, and the resulting statistical multiplexing can significantly increase throughput.

There are two distinct models for backboning legacy networks over ATM, namely classical IP (internet protocol) with address resolution protocol (ARP) over ATM, and LAN emulation (LANE). The first uses ATM as a high-bandwidth bit-carrier with quality of service (QoS) support, and the second supports virtual LANs, creating a bridged environment by hiding the underlying details of the ATM network from the users. Both of these models, as currently implemented, are subsets of the ATM subnet model, in which ATM is treated as a link-layer subnet (OSI terminology) and network layer addressing, such as IP, is preserved end-to-end. This contrasts to the peer model in which ATM is in effect elevated to the network layer and existing address structures (IP, IPX, etc) are translated to ATM NSAP addresses at the boundaries of the ATM cloud.

In the short term, the use of ATM within the futures test bed will encompass both the classical IP and LANE models, as well as their coexistence.

2.1 Classical IP and ARP

Initial local- and wide-area ATM (L-ATM, W-ATM) implementations draw on the classical model. From the standpoint of end-applications, ATM is another link-layer transport mechanism, peered with, for example, Ethernet, FDDI, and frame relay. Transit from one link-layer technology to another is accomplished through the use of routers (dedicated or workstation-based). Using IP as an example, Ethernet segments and FDDI rings are provided with subnet numbers, while the ATM backbone cloud is assigned a single subnet or a number of point-to-point subnets. Implementation of the classical model has no effect on the pre-existing logical infrastructure, as the subnetting scheme is preserved. Examples of this include the use of L-ATM as FDDI and W-ATM as frame relay replacements. The IP subnet originally dedicated to the campus FDDI backbone or the frame relay WAN may be reused within ATM. Despite its simplicity, the classical model provides capabilities beyond high bandwidth permanent and switched virtual circuits (PVCs and SVCs, respectively). As different users or applications may be mapped to VCs with differing QoSs, effective support of multiple real time data streams becomes a reality. Additionally, because of the ATM cell structure, latency and jitter may be minimised (or controlled) for those provided with resource reservation and flow-control mechanisms.

RFC 1577 provides the basis for the classical model, and specifies an ATM ARP server to be used for registration and resolution of Layer 3 protocol to ATM hardware addresses. Although the RFC is titled 'Classical IP and ARP over ATM', this architecture can be extended to other protocols. Routeing is primarily implemented with 'classical' bridging being supported by replicating the data to be bridged on each VC. However, this bridging option is not widely used due to both scalability concerns and the advent of LAN emulation. The ATM Forum's User Network Interface (UNI 3.1) standard addresses signalling issues as they relate to implementing IP over ATM, and RFC 1695 describes SNMP v2 managed objects for ATM interfaces, devices, networks, and services. This provides for the configuration, reconfiguration, and release of VCs over the ATM network. In keeping with simplicity, the interim local management interface (ILMI) is based on SNMP, though, in contrast to it, provides for a two-way exchange of management data. Parameters exchanged over the ILMI include the speed, media, type of interface (public or private), the number of operational VCs along with their quality of service, and address registration.

Routeing over the classical model takes place in the same way as routeing over any non-broadcast medium, but with one exception. When an end system wishes to send a datagram, the IP routeing table is first consulted for the next hop IP address. Note that the generation of this routeing table requires the periodic exchange of routeing data over the ATM network, most probably through the use of permanent virtual circuits (PVCs) in a full or partial mesh. After establishing the next hop, the system will check its ARP table to see if the next hop to ATM hardware address has been previously resolved. If not, an ATM ARP request is generated — the ARP server will respond with the mapping if the distant station has previously registered or will send a broadcast ARP on behalf of the sending station (Fig 1). With the distant ATM address now resolved, the end system may now signal an SVC to the destination.

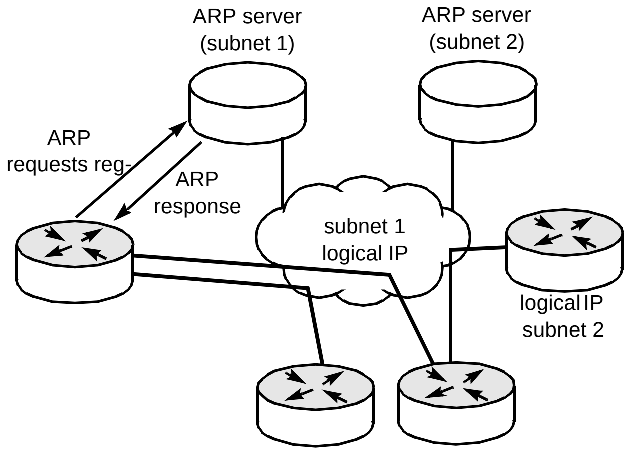


Fig 1 Operation of classical IP and ARP over ATM.

Implementations of the above model will consist of one or more work-group (WG), campus, or central office (CO) ATM switches at the core of a network supporting multiple logical IP subnets (LISs). In a large ATM network, this core may consist of a combination of WG switches in the local area and CO ATM switches as part of a public ATM service. ATM end systems will connect to these switches via ATM Forum (ATMF) UNI compliant interfaces supporting PVCs, SVCs, and traffic shaping. These end systems include routers, LAN switches with Layer 3 functionality, workstations, and PCs. All systems will implement a protocol stack supporting one or more of the common internetworking protocols such as IP, IPX, AppleTalk or SO (but, as described above, only IP systems are directly covered by RFC 1577). ATM ARP servers will exist on one or more ATM end system, within the network, with the number of server processes equalling the number of LISs in use. When an ATM ARP server is implemented on a switch management processor, accessible via SAR within the switch, it will appear as an ATM end system to ATM ARP clients. Finally, network management will provide for physical and logical control of the ATM network.

2.2 *Implementing classical IP*

The physical design of the network backbone was largely governed by the geographical locations of the users, and a design using six 16-port 155-Mbit/s interface switches was chosen (the costs involved in cabling meant that use of a single large switch was impractical). Single-mode fibre was laid between buildings, with multimode fibre laid for certain high-bandwidth routes within buildings. The initial network would not be fully meshed, but rather consist of 2 connected ‘stars’, although diverse physical routes will soon be provided.

Unshielded twisted pair cabling (UTP category 5 — BT’s ‘OSCA’) was laid in most of the buildings from each user to a central communications room. Although this would initially be used predominantly for 10 Mbit/s Ethernet, it was installed and tested for operation at speeds up to 155 Mbit/s, thereby permitting upgrade to either fast ethernet (100 Mbit/s) or direct ATM at 155 Mbit/s.

The first implementation phase (Fig 2) was of classical IP over ATM, and used 6 Cisco LS100 ATM switches and 8 Cisco 7000 routers.

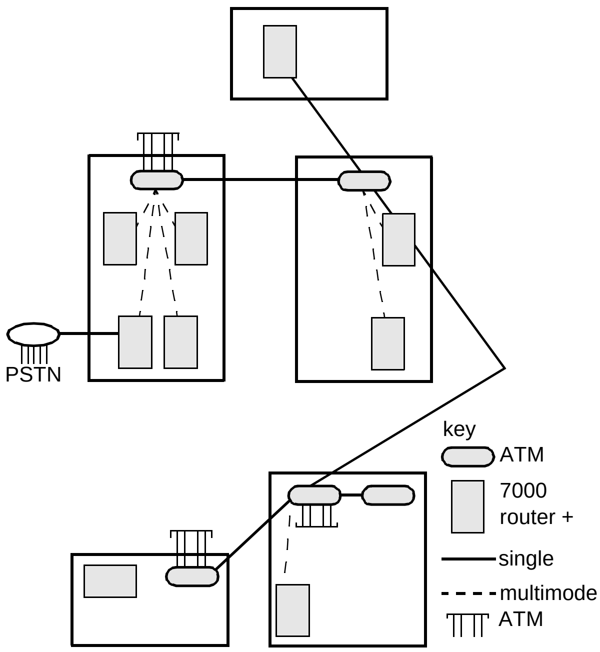


Fig 2 The network design for the first phase ‘classical IP’ implementation of the futures test bed.

- Cisco LS100 ATM switch

  The LS100 is a fully non-blocking, 2.4-Gbit/s input/output buffer type switch fabric with a minimum of 1000 cells of virtual output buffering per port. It can provide different output buffers for both variable and constant-bit rate traffic. It has two priority queues for cell delay — one for delay sensitive traffic and one for delay tolerant traffic. The switch supports permanent virtual circuits (PVCs), switched virtual circuits (SVCs), virtual channel (VCI), virtual path (VPI), point-to-point, and point-to-multipoint connections. The futures test bed has been designed using 155-Mbit/s cards using both single- and multi-mode optical fibre and unshielded twisted pair (UTP Cat 5) as transmission media, with support for both UNI 3.0 and 3.1.

- Cisco 7000 router with AIP ATM card

  The Cisco 7000 provides high-performance routeing and bridging. The system supports multiprotocol, multimedia routeing and bridging and any combination of Ethernet, Token Ring, FDDI, serial, HSSI (high-speed serial interface) and ATM media. The implementation in the futures test bed used AIP ATM interface processor cards for both mono- and multi-mode fibre, and Ethernet interface processor cards. This combination allowed 24 Ethernets to be linked to a single 155 Mbit/s pipe. Software on the router supports classical IP and ARP over ATM, to include an ATM ARP server process on one or more routers within an ATM domain, the ILMI, and both RFC-1483 encapsulation types (LLC/SNAP and VC-mux). Signalling for SVCs is based on ATM Forum UNI 3.0 or 3.1 (user selectable). Additional services supported over the UNI include 'classical' bridging based on RFC-1490, and LAN emulation (both as a LAN emulation client and as a LAN emulation server).

Other departments at BTL are served by a network of ethernets, connected via an FDDI backbone, and the users need continued connectivity to this network. Clearly, interworking between the new ATM network and the old network could be problematic, and the correct choice of IP numbers would be crucial. As the site legacy network primarily uses the RIP routeing protocol, it was decided at an early stage that the two networks would be linked at a single point to eliminate the possibility of unwanted loops. The new network would also have to use the EIGRP routeing protocol to enable variable length subnet masks to be used, since very fine subnets were required.

The phase 1 network using classical IP on Cisco 7000 routers was allocated 4 contiguous class 'B' subnets, giving a total of 1024 IP addresses. As each user requires their own ethernet subnet, a 255.255.255.252 subnet mask is used, which allocates four addresses to the user (only one of these is actually available to the user, as the others are required as broadcast or gateway addresses). So the first 200 users consumed some 800 of our 1024 addresses — not very efficient, but necessary. The remaining 200 or so addresses were used for subnetting the ATM backbone and for sundry other purposes (some users require more than a single IP address). This ATM backbone comprised a mesh of PVCs linking every 7000 router to every other through the LS100 ATM switches. Additional PVCs were created to link the first 'power users'— some Sun Sparc and Silicon Graphics Indigo workstations equipped with ATM network interface cards (NICs).

Because the legacy FDDI network uses RIP, the two networks were attached by static routes. In this way the RIP router process was fooled into thinking the futures test bed comprised four conventional 256-network segments. This combination of routers and ATM switching provided an integrated production ATM network, and served to demonstrate the use of ATM as a backbone technology within a 'classical' IP model, prior to the deployment of LAN emulation.

2.3 LAN Emulation (LANE)

The concept of virtual LANs allows the creation of what to end systems look like classical LANs but what are in fact combinations of various network technologies. LANE is defined by the LAN emulation group within the ATM Forum. It introduces a level of intelligence to the ATM network which allows us to create virtual Ethernet or Token Ring LANs with support for broadcast and multicast traffic.

LANE clients (LECs) are the interface between the classical LAN and the emulated LAN. Communication between clients is managed by the LAN emulation server (LES). The job of the server is to advertise the configurations of each client to the other clients so that transparent bridging between physical segments can be performed across the ATM network. In most cases, several VLANs will coexist in the same ATM-connected router or LAN switch.

For example, it might be decided that three departments in an organisation want to have their own logically separate virtual LANs. The 'finance', 'engineering' and 'human resources' VLANs would comprise a number of physical ethernet segments terminating on an ATM connected device running LAN emulation client software. Because of the physical locations of department members one device may terminate three physical 'human resource' segments, two 'engineering' segments and a 'finance' segment, another device may terminate two 'human resource' segments, one 'engineering' segment and five 'finance' segments and so on. Using the VLAN technology, interaction between all the clients and the LAN emulation server makes each department appear to be in its own bridged LAN and regular protocols, such as ARP, behave as would be expected in a bridged environment. If IP subnetwork addresses are used to differentiate the VLANs, a router can provide inter-VLAN connectivity.

During LANE initialisation, the LEC will first obtain configuration information from a LANE configuration server (LECS), which is then used to obtain the identity of the LANE Server (LES). The LEC will then register with this LES, which is the management entity within a single emulated LAN. When sending data, the LEC resolves a destination's MAC address to an ATM address using the services of the LES which knows which MAC addresses are on which physical segments and which LEC is responsible for it. Using the ATM address the LEC will open a unicast virtual circuit (VC) to the destination's LEC. All broadcast, multicast and traffic for destinations unknown or unresolved will be forwarded to the appropriate broadcast and unknown (BUS) or multicast servers, which would typically be physically collocated with the LES (Fig 3).

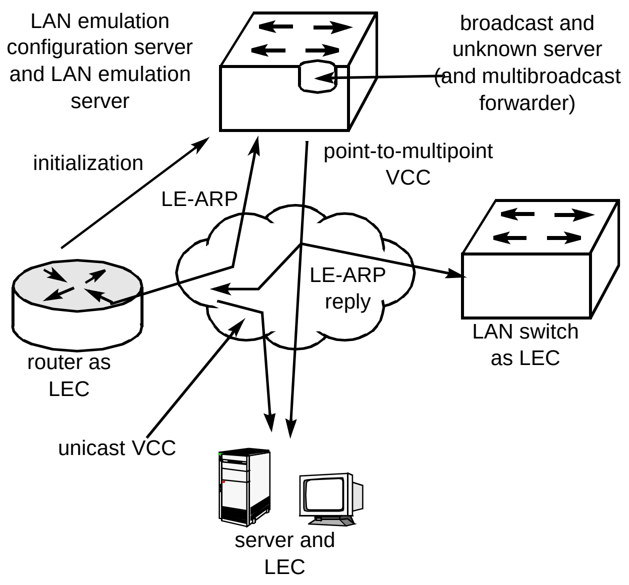


Fig 3 Operation of and devices associated with LAN emulation.

The LANE standard is by no means perfect. Currently it addresses only a single emulated LAN, and only allows one LAN technology to be emulated at a time — Ethernet or Token Ring, but not both. Also Layer 3 functionality such as IP is still required for the interconnection of emulated LANs even though they share a common ATM physical infrastructure, and ATM service features such as constant bit rate (CBR), variable bit rate (VBR) and available bit rate (ABR) are not available to LAN users. The LANE standard seeks simplicity by hiding the underlying ATM network. VCs are opened with an unspecified bit rate (UBR) which has implications for the sharing of network resources in the support of real time traffic under heavy network loading.

LAN emulation is sometimes considered as a possible W-ATM service offering. The service provider would install and operate centrally located servers (LECS, LES, BUS, M-cast), while the LECs would in most cases be located on the customer's premises. For security considerations it may be desirable to operate a separate physical LES for each customer connected to the service. An advantage of LANE over traditional LAN organisation is its support for mobility. From a purely physical standpoint, any users within a work-group should be able to access the emulated LAN for their work-group anywhere on a campus. Of course, though this may be technically possible it may not always be desirable. Due to problems with physical security and spoofing, a network administrator may want to limit access to a work-group's resources. Instead of universal access, a given LEC may only support pre-configured emulated LANs or may only support users with pre-configured MAC addresses. In the extreme, all mobility may be disabled and security is ensured at the expense of convenience.

LANE implementations include one or more ATM switches, as well as the necessary servers and clients. The LES may be implemented on any ATM end system such as a router, workstation, or PC, as well as on an ATM switch management processor as can the multicast, broadcast and unknown servers. LECs will be built into ATM end systems such as Ethernet and Token Ring switches equipped with ATM interfaces. They may also include Layer 3 functionality such as IP routeing. Where multiple emulated LANs must be supported over a single ATM infrastructure, or where connectivity to points beyond the emulated LAN is required, a router or a device with router functionality will be installed. As with the classical model, a network management system is also an important consideration, especially because of the need for logical work-group configuration.

In contrast to the classical model, LANE introduces a challenge for administrators of existing networks in that it de-couples addressing from topology and can now associate it with distributed users forming multiple work-groups. As each work-group forms a VLAN, using IP as an example, subnets originally assigned to ports on routers, often terminated on hubs, will be reassigned to this VLAN. If a single work-group occupies two or more physical interfaces on the router for capacity reasons, each of these interfaces will nevertheless be assigned the same subnet number. This is a change from most existing router implementations. As LANE is designed to satisfy the bandwidth requirements of LAN users, most LAN segments will be connected to ATM via LAN switches.

These devices will forward traffic within a work-group (Layer 2), but will often be provided with the capability of Layer 3 switching. This will allow traffic between work-groups to be forwarded directly from the LAN switch without passing through a central router. This router will still play an important part, though, in that it will act as a route server for these LAN switches. When forwarding data on Layers 2 and 3, these devices are often referred to as ‘multilayer switches’. Thus, a new device is introduced into the ATM environment, capable of supporting a great number of switched LANs at a relatively low price per port in comparison to routers. When implementing LANE in an existing networking environment, the network administrator will often begin by implementing the LES and LEC functions on existing network devices (routers, switches, and workstations). As bandwidth requirements increase, the administrator will install LAN switches acting as LECs.

It is possible to implement both models over a single infrastructure in the following way. Assume an IP environment such as the futures test bed, where ATM was first implemented as a simple backbone. The installed base consists of routers, managed hubs or LAN switches, and one or more ATM switches. At this stage a traditional IP subnetting structure is used and one router provides connectivity to WAN services. This same router will act as the RFC 1577 ATM ARP server. Upon implementation of LANE, a LES function will be added to this same router, with a subset of the router LAN interfaces associated with one or more VLANs. Each of the remaining routers will act as a LEC client for directly attached LAN segments as well as devices connected to downstream hubs and LAN switches (switches implementing transparent bridging and connected to the router via an FDDI interface, for example). The router provides for connectivity between the classical and LANE domains.

Because of the later standardisation of LANE (end of 1994) against the standardisation of the classical model, products supporting LANE, and especially those implementing support for the interconnection of multiple VLANs and route distribution, are less mature from both a hardware and software standpoint. The first L-ATM products — work-group switches and workstation interfaces — were released in 1993. These pre-UNI 3.0 products made use of PVCs or SVCs based on proprietary mechanisms. No automatic configuration via the ILMI or address resolution based on the ATM ARP server existed at the time. Standards-compliant SVCs had to wait until the end of 1994, when multi-vendor support for the ATMF UNI 3.0 (SVCs) and the interim inter-switch signalling protocol (IISP) first appeared. At this same time, vendors announced ‘beta’ implementations of the ATMF LANE protocol on switches, LAN switches, workstation interfaces, and routers. This included LEC and LES software as well as hardware consisting of ATM-equipped LAN switches. Thus, by the beginning of 1995 it became possible in principle to implement either model.

2.4 Implementing LANE

Figure 4 shows a diagram of the completed futures test bed network which embodies both LANE and wide area switching. This utilises a 7000 series router to provide LES functions with Catalyst 5000s at the edges of the network.

- Cisco Catalyst 5000 Ethernet switch/ATM concentrator

  The Catalyst 5000 is a highly-optimised ATM-capable LAN switch. It is a modular switching system, initially equipped with five slots and a 1.2 Gbit/s frame backplane. For the futures test bed, a combination of 12-port 10/100BaseT and 24-port 10BaseT Ethernet modules are being used. Additional interfaces included FDDI, 100 VG-AnyLAN, combined Ethernet switching and hubbing, and Token Ring. The Catalyst 5000 software includes both LES and LEC support, as well as Layer 3 switching for interconnection of VLANs.

As the 7000 Series also supports LAN Emulation, the completed network consists of a combination of Catalyst 5000s and 7000s connected to the ATM switches. One 7000 is acting as the Layer 3 interworking unit between VLANs, as well as acting as an edge system between the futures test bed LAN emulation environment and wide area ATM services based on the classical model (Fig 4).

This switched Ethernet/ATM solution is likely to find wide applications with many business customers over the next few years.

3. Test bed applications and concerns

The futures test bed brings together legacy networks, state-of-the-art ATM and tomorrow’s applications and approaches to work. Test bed users are increasingly dependent upon their networked computers for every aspect of their work:

- they communicate using e-mail, commonly exchanging multi-megabyte attachments,
- they publish information on themselves and their projects using the World-Wide Web,
- they develop software to search these servers for relevant information and produce summaries of what they find,
- they use high quality video conferencing and virtual environments to link physically separate locations.

Yet the test bed is not an isolated ATM island — remote users, whether working from home or on the move, enjoy secure transparent network access via dial-up modems and

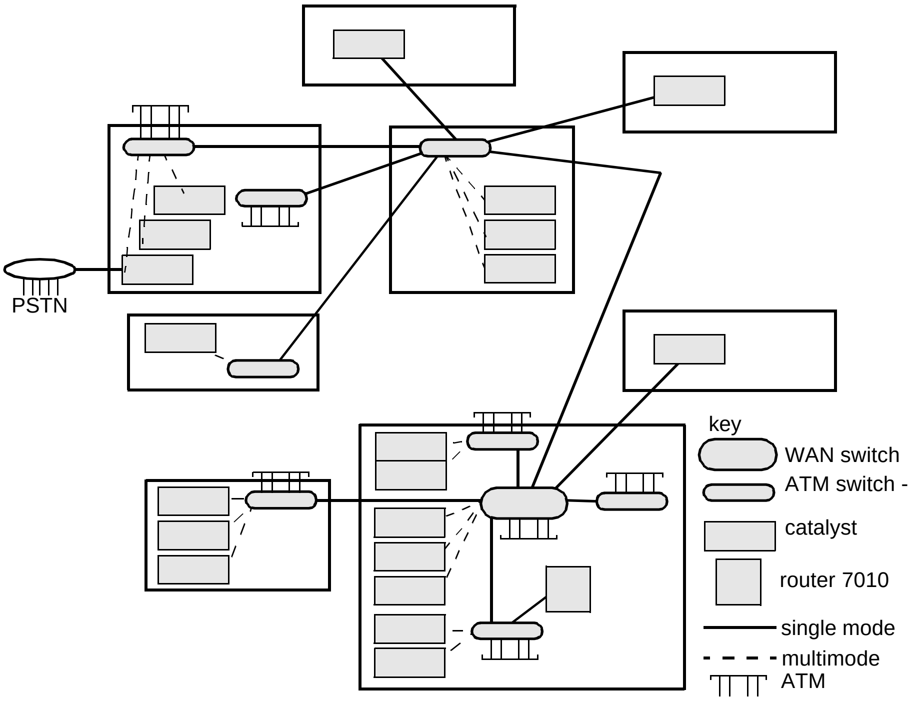


Fig 4 Test bed phase 2, including LANE implementation, and wide area networking (WAN).

ISDN. Although these users lack access speed, many applications substitute processing inside the fast network for bandwidth over the wide area, so that some of the benefits of ATM are available even over dial-up lines. Higher bandwidth connections to other sites also facilitate ongoing collaborative work with remote R&D teams.

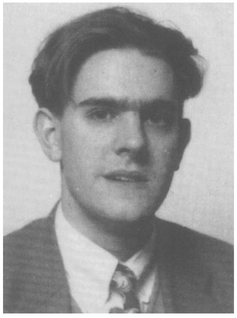
The operation of the ATM network is of prime importance to all of these applications, but it is not intended to protect it from the users. Rather, the early adoption of new ATM equipment and standards in co-operation with Cisco and other suppliers is being encouraged. By using ATM in a demanding live environment, it is hoped to gain experience and understanding of potential problems before they manifest themselves in public networks. Some areas of particular interest include:

- performance of legacy protocols over ATM — whether LANE is just an expensive way of replacing coaxial cable and increasing its latency,
- evolutionary issues for the TCP/IP protocol suite — whether protocols designed for unreliable networks can ever operate efficiently over ATM, and to what extent performance must be sacrificed for interconnectivity,
- mapping of higher-level resource reservation to ATM,
- whether ATM networks can ever support the future's highly delay-sensitive traffic for communications media beyond simple sound and vision,
- how well connection admission control schemes work with bursty traffic and indiscriminate loading (e.g. TV phone-ins) in comparison with traditional lost calls cleared telephone networks,
- how ATM networks should be managed and administered, particularly with respect to large-scale public networks.

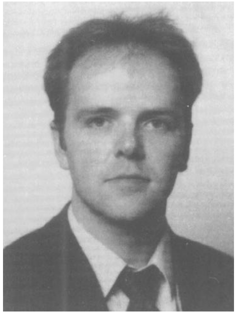
In conclusion, the futures test bed at BT Laboratories is an exciting initiative embracing ATM and its bandwidth without compromising connectivity. A demanding community of users is reliant upon the network for their daily work, and they are pushing the technology to its limits.

ATM may be central to the public networks of the future, but there are risks associated with shortening development life cycles and the flattening and decentralisation of network topologies. Traffic is becoming increasingly bursty and broadband, and today's LANs provide a good model of the future. The tendency to increase switching and processing within the LAN, and the consequent increases in latency, which may lead to uncompetitive performance in the future information economy, is also noted.

The test bed and its users are a living experiment. Its foundation is state-of-the-art ATM, but the users are ensuring that it continues to evolve as they probe the future. In the next few years, the inclusion of wavelength division multiplexing and switching will add a further dimension for the future.

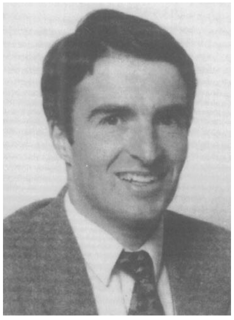


Jeremy Barnes received a degree in Physics from Durham University in 1991. He joined the advanced modelling unit at BT Laboratories, investigating demand systems for broadband networks and services. Over the past two years, he has worked on analysing information networking technologies and building understanding of the emerging information market. His current work involves the practical realisation of information networking concepts over a high-bandwidth asynchronous transfer mode (ATM) campus network.

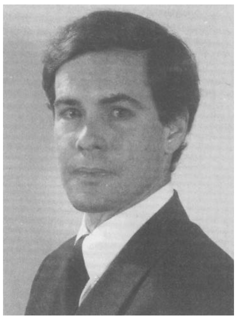


Jon Chalmers began work with BT Laboratories in 1989 after graduating with a degree in Computer Science from Queen Mary College, London. His work has included software development for an office automation system and for network management.

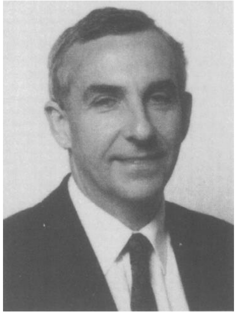
He is now a senior member of the team researching distributed applications over high-speed data networks and has a particular interest in messaging.



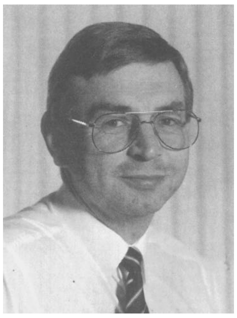
Dave Newson graduated from Cambridge University with BA and PhD degrees in Physics. He spent two years working in the R&D Center of Toshiba in Kawasaki, Japan, researching optical phenomena in III-V semiconductors. After joining BT he worked on the development of optoelctronic and millimetre wave electronic components. Part of this work was directed towards systems requirements for ATM. He then joined a team to design, install, commission and run the ‘futures test bed’ ATM-backboned network at BT Laboratories. He has been author of many papers and three patents.



David Ginsburg is employed by Cisco Systems Europe as an internetworking consultant. His current focus is on ATM deployment, marketing, and standardisation. He graduated with a BSc in Electrical Engineering from Renssalaer Polytechnic Institute in 1985. Until joining Cisco Systems, he was employed by Alcatel SEL as part of the metropolitan area network project management team, working with the German Telecomms on SMDS deployment. Earlier work included the design of survivable and secure internetworks within the US Government.

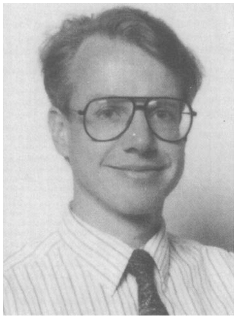


Ian Henning attended the University at Cardiff where he gained a BSc (Hons) in Applied Physics. He remained there for a further four years, three of which were spent gaining a PhD on Defect Levels in GaAsP LEDs, and the fourth as a fellow extending this work on to InGaAsP LEDs and lasers, and InGaAs photodetectors. In 1989 he joined BT Laboratories and began working on the design, modelling and characterisation of all aspects of semiconductor laser performance. Since then he has led a group whose responsibilities have covered the design, fabrication and characterisation of optoelectronic integrated circuits and photonic integrated circuits. Latterly this work emcompassed the development of novel components for interfacing between radio and optics. In April 1994 he changed job to head up a group which is responsible for designing, building and running an advanced ATM-based broadband network, to provide broadband services to the desktop for 650 users. He is currently a Visiting Professor at the Department of Electrical Engineering, Sheffield University.



Peter Cochrane joined BT in 1962. He graduated from Trent Polytechnic with a BSc Hons in Electrical Engineering in 1973 and gained an MSc, PhD and DSc in telecommunications systems from the University of Essex in 1976, 1979 and 1993 respectively. He is a fellow of both the IEE and IEEE, a visiting professor to Essex and Southampton Universities, an honorary professor at the University of Kent, and a visiting fellow to University College, North Wales at Bangor. He joined BT Laboratories in 1973 and has worked on a variety of analogue and digital switching and transmission studies.

In 1978 he became involved with the development of intensity modulated and coherent optical systems, photonic amplifiers and wavelength routed networks for terrestrial and undersea applications. In 1991 he was appointed to head the systems research division at BTL which is concerned with future advanced media, computing and communications developments. He is currently head of advanced applications and technologies at BTL.



Dave Pratt has a PhD in lasers and nonlinear optics, and began his career with fibre optics at Plessey Research, Roke Manor and since 1987 at BT Laboratories. During this time he published many papers and had seven patent submissions.

After a career change to networking, he is now a senior member of a team researching advanced networks. He is jointly responsible for deploying and configuring the ‘futures test bed’ campus ATM network.